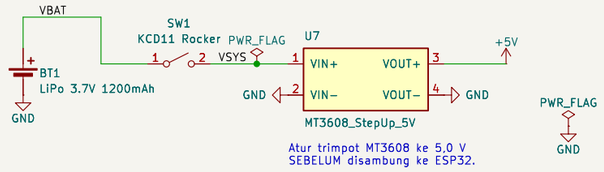
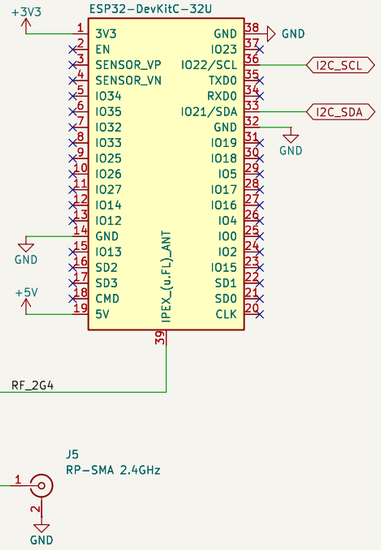
A. SUPLAI DAYA (Li-Po -> Saklar -> MT3608 5V, tanpa charger)

Tanpa modul charger: Li-Po dicas di luar alat
(charger Li-Po 1S 4.2 V). Pakai baterai dengan PCM
internal (proteksi over-discharge/short).

Matikan SW1 saat ESP32
disambung ke USB (flashing).

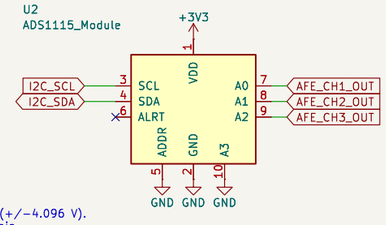


B. MIKROKONTROLER & RF

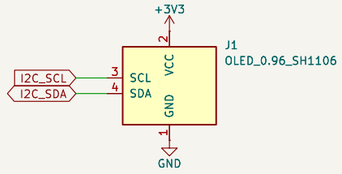


U1: ESP32-DevKitC V4 dengan ESP32-WROOM-32U.
J5: soket RP-SMA bulkhead + antena 2.4 GHz,
disambung ke IPEX/u.FL U1 lewat pigtail RG1.13.
I2C: SDA = GPIO21, SCL = GPIO22 (main.cpp).

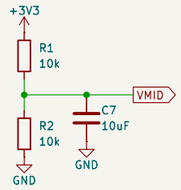
C. ADC 16-BIT, OLED & REFERENSI VMID



ADDR=GND -> 0x48, GAIN_ONE (+/-4.096 V).
A0 Glukosa, A1 pH, A2 Fruktosamin.

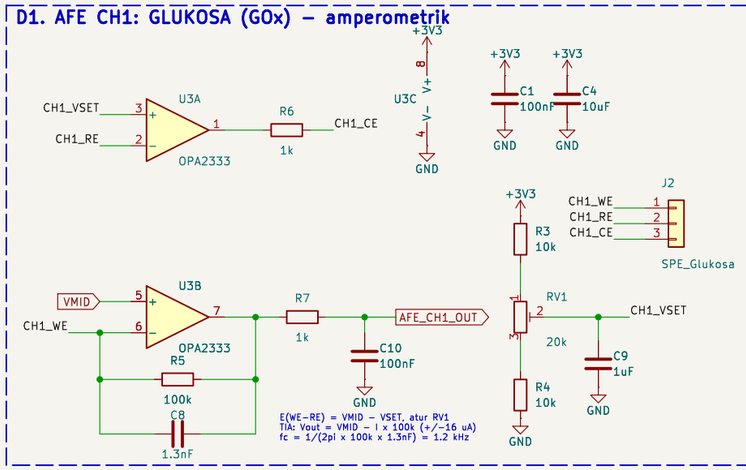


OLED SH1106G 128x64, alamat 0x3C.
Cek urutan pin modul (ada varian VCC-GND).



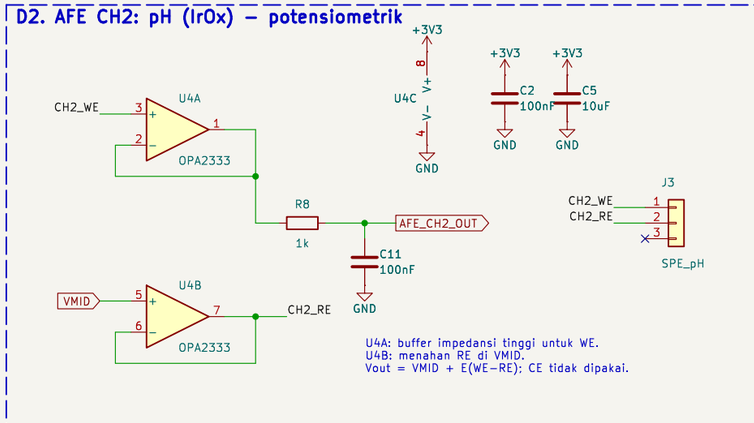
VMID = 1.65 V: potensial WE (CH1, CH3)
dan RE (CH2).

D1. AFE CH1: GLUKOSA (GOx) - amperometrik



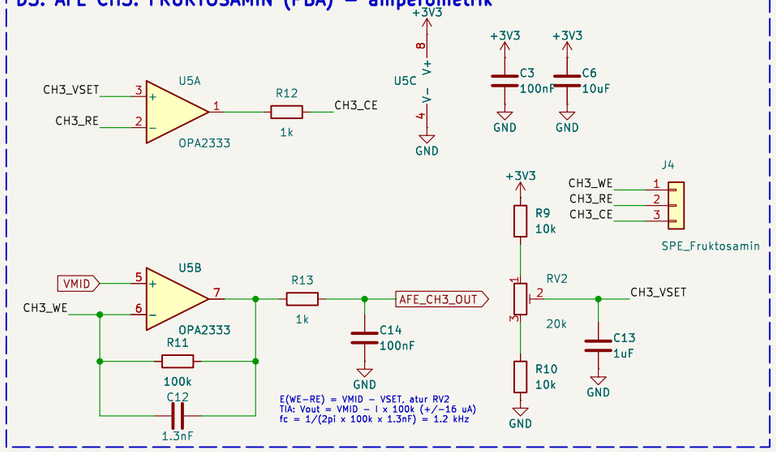
E(WE-RE) = VMID - VSET, atur RV1
TIA: Vout = VMID - I x 100k (+/-16 uA)
fc = 1/(2pi x 100k x 1.3nF) = 1.2 kHz

D2. AFE CH2: pH (IrOx) - potensiometrik



U4A: buffer impedansi tinggi untuk WE.
U4B: menahan RE di VMID.
Vout = VMID + E(WE-RE); CE tidak dipakai.

D3. AFE CH3: FRUKTOSAMIN (PBA) - amperometrik



E(WE-RE) = VMID - VSET, atur RV2
TIA: Vout = VMID - I x 100k (+/-16 uA)
fc = 1/(2pi x 100k x 1.3nF) = 1.2 kHz

CATATAN DESAIN
1. Topologi AFE dan nilai R/C adalah rancangan awal (repo belum punya data rangkaian AFE aktual). Wajib diverifikasi dengan uji bench dan kalibrasi.
2. Rf 100k + Cf 1.3nF dipilih agar bandwidth TIA ~1.2 kHz sesuai simulasi di laporan PKM. Ganti Rf bila arus sensor > 16 uA.
3. Modul ESP32, ADS1115, MT3608, OLED ada di library proyek Glicovia.kicad_sym. Pull-up I2C 10k sudah ada di modul ADS1115.
4. Skematik ini digenerate oleh generate_schematic.py; edit skrip lalu jalankan ulang agar netlist tetap terverifikasi (verify_netlist.py).

AFE: rancangan awal, perlu verifikasi bench
Daya: Li-Po 1S (PCM) -> KCD11 -> MT3608 5V, tanpa TP4056
ESP32-DevKitC + ADS1115 + 3x OPA2333 potensiostat + OLED SH1106
Tim PKM-KC Glicovia

Sheet: /
File: PKM_Glicovia_TanpaTP4056.kicad_sch

Title: PKM-Glicovia Handheld Biosensor - Tanpa TP4056

Size: A3	Date: 2026-10-18	**Rev: 1.0**
KiCad E.D.A. 8.0.9		Id: 1/1